5. *No discrimination against persons or groups:* An open source programming language must not discriminate against any kind of person or group of persons.
6. *No discrimination against fields of endeavour:* An open source programming language must not restrict anyone from making use of the program in a specific field of endeavour.
7. *Distribution of licence:* The rights attached to the open source programming language must apply to all to whom it is distributed, without the need for executing additional rights by those parties.
8. *The licence must not be specific to a product:* The rights attached to the open source programming language must not depend on it being part of a particular software distribution. If it is extracted from that distribution and used or distributed within the terms of its licence, all parties to whom it has been redistributed should have the same rights as those that are granted along with the original programming language distribution.
9. *The licence must not restrict other software:* An open source programming language must not place restrictions on other programming languages that are distributed along with the licensed programming languages.
10. *The licence must be technology-neutral:* There is no provision for the open source programming language to be predicated on any individual technology or style of interface.

Open source programming languages in the market

A flashback to the 18th century will reveal that the world's very first program was written by Ada Lovelace for calculating Bernoulli's Number using the Analytical Engine. The culture of writing programs has come into existence since then and, subsequently, it has led to the development of various other programs that are used to perform some complex mathematical calculations. According to Wikipedia, it was only in the 1950s that the first high-level programming language, *Plankalkül,* was designed and developed by the Germans to communicate instructions to the computer. John Mauchly's Short Code was the first high-level language ever to be developed for an electronic computer.

Today, developers and programmers have many programming language options that can be used to develop applications of their choice. Let's take a look at a few such popular open source programming languages.

1. Java

Java is one of the world's most influential programming languages developed so far, and it is mostly open source today. It is being used at the core of many Web and Windows based applications on all platforms, operating systems and devices. This class-based object-oriented programming language has a large number of features.

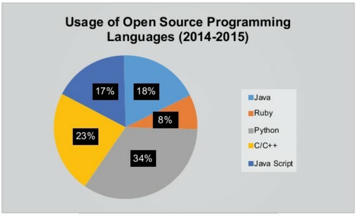

Figure 1: Open source programming languages in use (*Data source*: Lifehacker community)

Features of Java

- Is object-oriented.
- Allows us to create various modular programs and reusable code.
- It is easily ported.
- Is platform-independent.
- Easy to write, compile and debug.

2. PHP

PHP is on its way to becoming the most popular open source programming language. According to many leading industry leaders, PHP has emerged as the most user-friendly open source language; therefore, various open source packages such as Joomla and Drupal are built on it. It's even budget-friendly and, hence, PHP based solutions are being used by entrepreneurs and SMEs as well. Currently, many developers are making their debut on PHP, which clearly highlights its strong community base.

Features of PHP

- Cross-platform compatibility.
- No need to specify the data type for variable declarations, and predefined variables can be used.
- Availability of predefined error reporting constants.
- Supports extended regular expressions.
- Has the ability to generate dynamic page content.
- Allows the user to create, open, read, write, delete and close files on the server
- Output can be in HTML, images, PDF, Flash, XHTML and XML file formats.
- Runs on various platforms such as Windows, Linux, UNIX, Mac OSx, etc.
- It is compatible with almost all servers being used currently.

3. Python

Python was developed by Guido Van Rossum in the 1980s and handed over to the non-profit Python Software Foundation, which now serves as the administrator of the Python language. It was one of the first programming